LECTURE 01 :

Python ka Chilla :

python

We learn python with baba ammar next 40 days.

is sa data science ka algorithm bnana start sa end tak kry ga.us ka liye ma apka instructor hun **self taught data scientist**.

it will be a greate fun to learn.very culture or in desi way .

is ka liye hum language jo use kry ga wo python hai.python aik programming language hai or ya itny easy hai ka agr ap ko english aty hai tou ap pythonn seekh jaye ga.no prior knowlwdge required for learning this course.pyhton ka liye koi age limit nai hai.

is ma collaboraters jo hein;

1. codanics
2. schorlarship network

certificate nai hai, no charges, if you are anywhere you can join this course.

real life problem solved within course.

DATA SCIENCE

data science future hai agarr ap kbhy khy pa b mobile pa ya google pa ya ticket ki booking ka waqt activity krty hein ap tons of data produce krty hein tou is data sa decesion making krna data sceine khlata hai.kiu ka ap jab data provide krty hein tou ap apna virtual apperance internet pa da rhy hoty hein. jis sa hum data retrieve krny ka ba bad further process kar skty hein.

1. kuch airlines predivction dety hein ka hmary flight delay hai that was bcz data science.
2. environmental department b yhi se krty hein.
3. amazon daraz google facebook sabhy yhi use krty heun.

bioinformatics sa ap genes ka pata lga skty hein this is also data science.covid vaceine ka bad log khty hein 5% logo ko dil ki bimari hoty hai ya b data science sa hi pata lga skty hein.

research hypothesis testing kha pa ap apna research hypothesis jo usy test kar skty hein.pory ka pora data analysis python ka upar kar skty hein or data sceince pa kar skty hein or is ka liye ap hypothesis kesa bnana hai isy test kesy krna hai data visualize kesy krna hai data deriven decesion making kesy krny kesy hai result ko kesy likhna hai or top class ki visualization graph kesy bnany hein ya sab hum is course ma dekhy ga.

why should we use python

the most popular language in the world is python.or jo log is pa kam kar rhy wo janty hein is ka kia impact hai.

kon kon isy use krty hein

- software engineers
- sceintists
- data analyist
- mathematicians
- statisticans
- accountants
- network engineers
- everybody can use python

  python is a future.you don't need to be A IT guy.ya open source hai.ziada tar log jo upser cheezy mention in ka lia python hi use krty hein.data sceince with python is more benefiical then R.data sceince is the most sexiest job in 2021.python is scaleable means ka aik dafa script likha or usy copy b kar skty hein ya chly ga.

ya kha kha use hoty han :

1. data analysis
2. AI AND ML
3. web apps
4. mobile apps
5. desktop apps
6. software testing
7. HACKING (ETHICAL HACKING)

python is a 36 year old language firstly it developed in 1980.creator of python is Guido Van Rossum (dutch programmer in netherland).

LECTURE 04 :

installation & first line of code

window ka start ma ja ka command prompt(cmd) open karky likhy **python --version**

python ko hum run tbhy kar skty hein jab hum isky syntax ka mutabik code likhy ga agr hum isy apny language ma command dein ka 5 ko 5 ma add kar do tou ya error da ga so humy syntax ka according command deny hai like

```
print("hello, world")

print(2+3)

print("that was my first program")
```

is code mein hum na print use kia jo python mein output ka liye istemal hota hai. is code mein strings hein.collection of words ko double quotes mein likthy hein inhy strings khty hein or is code mein mathematical operations istemal hovy hein

cmd ma dir likhy tou desktop ki sary directories ki list open ho jaye gi.

cmd ma cd desktop likhy tou desktop pa chla jein ga.

agar hum new dir bnana chaye tou **mkdir** likh kar agy jo nam rkhn hai wo likh dein.

phir agr ap folder dhkna chahty hein tou cd ka agy folder ka address likhy.agar folder pechy jana ho tou likhy **cd..** or agar agy jana ho tou **cd.**

visual studio code is an IDE.python ki file kaa lie .py hamesha likhy.zoom in ka liye ctrl++ kry.

LECTURE 05

complete python tutorial for biggner

1-OPERATORS

operators ko run krny ka kfy tariqy hoty hein yha par hum dekhy ga terminal mein operator kesy run krty hein

- sab sa phla operator addition(+) ka hai.we can write in cmd for use of addition operator.

  ```
  print(2+2)
  ```

- second operator subtraction ka hai. isy istemal krny ka liiye cmd ma likhne pry ga.

  ```
  print(3-2)
  ```

- third operator multiplication ka hai. values ko multiply krny ka liye hum histeric ka sign use krty hein like .

  ```
  print(3*2)
  ```

- forth operator division ka hai.values ko divide krny k liye istemal hota hai like we have to write for dividing two values it will show values in like float values.

  ```
  print(9/2)
  ```

- fifth operator is floor division(//). In this operator after dividing num is round off to smaller num.

  ```
  print(10//3)
  ```

- sixth operator is modulus or remainder.

  ```
  print(9%2)
  ```

  in this operator it will simplify like 2*4=8 but the remainder that is remaining is 1.which means the number left over after divide.

- seventh operator is exponent(**).this operator is used to give power to number.

  ```
  print(2**3)
  ```

spaces are meaning less in python.

PEMDAS(paranthesis, exponenece, multiply, divide, add, subtract) is sequence sa python code run krta hai.it will work left to right.M D & A S

2-STRINGS

koi b text jo excel ma use ho within single comma,double quotation mark ya 3 quotation mark usko strings bolty hein.

```
print("hello, world")

print("that was my first program")

print('thanks god')

print("actually i'm enjoying it")
```

```
print('''yes i know that it is multi line string i just wanna practice it''')
```

agar ap spaces string ka bhir lgae ga tou koi impact nai pry ga lkin agr ap sapces andar lgae ga tou wo exactly output ma b show hon gi.

3-comments

program ma jo code b hum run nahi krwana chahty usy hum comment out kar da ga comment out krny ka 2 tariqy hein aik **#** lga ka dusra **ctrl+/**. e.g

```
#print(2+5)

print("for comment") isy select kar ka ctrl+/ dba dein
```

it will work in up to down.

variables

variables object hoty hein containing spercific values.agar hum x=5 likhy tou x variable hai or is ki value 5 haii.ya aik integer variable hia.isy trha agr hum y="this is string" is mein y variable hai or is ka samny is ki value assign ki hoi hai ya aik string variable hai.

```
x=8

print(x)

y="uzair"

print(y)

x=6

print(x)

y=x+4

print(y)
```

types/class of variable

type of variable check krny ka liye print likh ka is andar type(variable) likhay ga like **print(type(x))** tou is ki class a jaye gi

rules to assign a variable

1- the variable name should not start with a number

2-the variable name should not contain special character we can only use characters or numbers or underscores.

3- the keywords cannot used as variables

4- do not space between variables.

5- short or descripted

6- python is a case sensitive language so be careful while assigning variables.

```
fruit_basket="mangoes"

print(fruit_basket)

type(print(fruit_basket))
```

is code mein fruit_basket aik variable jis mein mangoes value assign ki gai hai.nechy output ka liye print ka function use kia howa hai.3 line mein variable ki type/class check krny ka liye code likha gia hai.is ki class string aye gi.

4-input variables

user sa sawal pochny ka liye hum **input function** use kkrty hein like ma program bnata hun apko fruit konsa pasnd hai khud user ya customer choose kar la ga.

simple input function

```
fruit_basket="mangoes"

print(fruit_basket)

fruit_basket=input("which is your favourite fruit? ")

print(fruit_basket)
```

is code ma addition sirf input fuction ki hovy hai websites mein b esy tariqy sa sawal pochty hein.

input function of 2 stage

```
name=input("what is your name? ")

greetings="hello"
```

```
print(greetings, name )
```

is program ma phly hum na name variable rkha values mein input fuction use kia gia hai jis main nam pochha gia hai . second line mein greetings variable or agy string value . third line mein print fuction use kar ky bataia gia hai k phly greetings print kry phir name.

third stage input function

```
name=input("what is your name? ")

age=input("what is your age? ")

greetings="hello!"

print(greetings, name, ",you are still young " )
```

basically ya bi same hi hai bas is ma 3 input perimeters diye hein jis mein aik mein name pochta hai 2sra ma age pochta hai or 3sry mein greetings hai.print ma in ki sequence ya di hai ka phly greetings phir nam phir age or jab user age btaye ga tou ya us ki jaga pa you ARE STILL YOUNG likh da ga.

5-conditional logical operator

logical operators are either "true or false" or "yes or no" or "1 or 0"

- equal to ==
- not equal to !=
- less than <
- greater than >
- less than or equal to <=
- greater than or equal to >=

agar hum python sa pochty hein 4 equal to 4 tou wo error da ga pyton sa us ka syntax ka mutabik question pochty hein like

```
print(4==4)

print(4!=4)

print(4>3)

print(3>6)

print(5<6)

print(3<=5)
```

application of logical operators

```
ali_age=4

age_at_school=5

print(ali_age==age_at_school)
```

ya program bnia hai jis mein bchy ki age add kar ka pata kar skty hrin ka wo school ka liye eligible b hai ya nai.phly line ma bchy ki age btaty hein dusry line ma school ka liye required age third line ma equal ka operator laga ka pata kia jta hai ka bchy ki age school jany ka qabil hai ya nai.

hum jab netflix pa ya khy pa b movies waqera dhkty hein wha pa age ka pochty hein wo b esy method sa hoty hein

another method

```
hammad_age=input("how old is hammad? ")

hammad_age=int(hammad_age)

age_at_school=5

print(hammad_age==age_at_school)
```

is program ma input function or logical operator or type conversion istemal kiye gye hein

6-type conversion

aik data type ko dusry data type ma convert krny ko type conversion khty hein

```
x=10

y=10.2

z="uzair"
```

```
print(type(x))

print(type(y))

print(type(z))
```

is code mein x integer hai y float hai or z string hai

implicit type conversion

```
print(x*y)
```

implicit ma python khud convert kar leta hai or

agar hum x ko y sa multiply kren class float ma aye gi jab b integer ko float sa multiply ya plus kren ga tou class float mein aye gi

explicit type conversion

```
age=input("what is your age? ")

#age=int(age)

print(age, type(int(age)))
```

explicit mein hunmy khud krny hoty hai

phly line mein hum na input function lga ka user ki age pochy or dusry code mein hum na age ki class change ki hai or 3ry line ma hum na 2 sry line ko comment out kar ky or method sa class change ki

7-if else & elif statements

IF STATEMENT

```
hammad_age=2

required_age=5

if hammad_age==required_age:

print("hammad can join school")
```

is code ki phly line ma age btaye hai dusry ma required age btaye hai 3sry ma if statement or equal ka operator istemal karky btaia ya pocha gia hia ka agr age, required age ka equal hia tou print kry hammad can join school .

if else

jab ap if satement likhty hein tou ya b btaye ka gr ya statement true na ho tou ya print kar dy usky liye hum **if else** ka use krty hein. like

```
hammad_age=2

required_age=5

if hammad_age==required_age:

print("hammad can join school")

else

print("hammad can not go tto school")
```

elif statement

```
hammad_age=2

required_age=5

if hammad_age==required_age:

print("hammad can join school")

elif hammad_age>required_age

print("hammad should join higher school")

else

print("hammad can not join school")
```

elif mein hum bas multiple conditions use kr skty hein

8-functions

basiclly funtion **reuseable block of code** hota hai

agar program mein koi kam bar bar krna ho tou us ka function bna lo is sa ap ko bar bar code nai likhna pry ga blky ap jab b function ko call kro ga function chal jaye g

defining a function

def print_codanics()

> print("we are learning with uzair")
>
> print("we are learning with uzair")
>
> print("we are learning with uzair")

is line mein 3 cheezy hein phly or print ma wo data hai jisy hum function bnana chahty hein

- **def** --> define krny ka liye sab sa phly likhty hein ya aik python ki command hai
- **print_codanics** --> ya funtion ka nam hai
- **( )** --> ya paremeters function ka liye lazi hai

> print_codanics

ab hum jab b print codanics print kry ga tou ya function clhy ga

another way of defining a function

> def print_codanics():
>
> text="we are learning with uzair"
>
> ```
>     print(text)
>
>     print(text)
>
>     print(text)
> ```
>
> print_codanics()

third way of defining a function

> def print_codanics(text):
>
> ```
>   print(text)
>
>   print(text)
>
>   print(text)
> ```
>
> print_codanics("we are learning with uzair")

defininig a function with if,else & elif statement

> def school_calculator(age):
>
> if age<=5:
>
> ```
>   print( "you can join school" )
> ```
>
> else:
>
> ```
>   print( "you are not old enough for school")
> ```
>
> school_calculator(3)

is program ma phly line ma def ka function use howa hai baqi program as usual or end ma def ka function ko input dia hai jis pa process kar ka us na output da dia

9-loops

Loop ka matlab hai ek hi code ko baar-baar chalana (repeat karna)

jesa ka ap sabhy janty hein ka jab aik bcha 5 sa 10 saal ka hota hai tou uski mama usy rozana subha utha ka school bhjty hein until jab tak wo khud tyar ho kar jany ka qabil na ho jaye

2 trha ka loops hein

- while loop
- for loop

1. **while loop**

Jab tak condition True ho, loop chalta rahega.while loop ko infinite loop b khty hein

```
x=0

while (x<=5):

print(x)

x=x+1
```

phly line ma x=0 define kia or while function ko use kia jia ma conditional logical operator use kia ka jab tak x 5 ka less than ya equal hai tou x print hota rhy phir print ka functiona use kar ky btia ka x ko print kry or forth line ma increment opertOR USE KIA

- agr ya na krty tou x ki value infinite chlty rhty is liye while loop ma increment or decrement operator ko lazmi use krna chyie

2. **for loop**

Jab tak condition True ho, loop chalta rahega

```
    for x in range(6, 13):
        print(x)
```

jo range di gai hai jab tak is range ma value ayein gi loop chlta rhy ga for example

```
days=["mon","tues","wed","thur,"fri","sat","sun"]

for d in days:

if(d=="fri"):break

print(d)
```

hum na aik array bnai or dusry line ma isy d ka nam da dia 3 line ma hum na logical operator use kia or us ka bad break lga di us ka bad print d kar dia

10-import libraries

- ***bhut saray programmers hai jinhu na bhut sary functions define kiye hovy hein tou khud sa function define krny ki bajye open source ap unky fucntion use ka skty hein for example agar ap chahty hein ka pi ki value print kry tou hum math ki library import krty heun import krny ka baad math ka sary function hum use kar skty hein***

```
>import math
>
>print("the value of pi", math.pi)

>import statistics
>
>x=[150, 250, 350, 450]
>
>print(statistics.mean(x))
```

agar hum ya command dein ga tou iska mean ajye ga is trha sa hum different type ka statistics ki libraires ko use krsty hein is ma kia kia hai hum statistic. kar ka tab dabaye ga tou saby options a jaye ga

important libraries

- numpy
- pandas

11-trouble_shooting

trouble shooting is the process of finding and fixing errors in the program there are three main types of error

- syntax error
- run time error
- logical error

1. **syntax error**

agar hum language ka syntax ka according code nai likhy tou ya syntax error ho ga like

```
print("missing commas)
```

missing commas

2. **run time error**

program run krty waqt error aye like when we divide something to zero mathe matically error ko run time error keh akty hein

```
print(25/0)
```

3. **logical error**

poor knowlege of programmer like

```
name="uzair"

print("hello uzair")
```

ia error main user ki apny ghlty hoty hai.ya error sab sa difficult hai trouble shooting ma

---

LECTURE 06

saving your python code in digital notebook :

sab sa phly anaconda navigator ko install kry phir usy open kry open krny ka bad jupyter notebook open kry.notebook ma aia folder bnaye.python kernel ko use karky new file ntebook ma bnaye.ap kisy ko khty hein ya kha lo mera kaam kar do isy principal pa kernel ko khty hein ya mera kaam kar do .

kernel wo chz hoty hai jo usy run krta hai, execute krta hai then give you output.new notebook bna lein notebook pa ap file bna lein.or is ka layout ma cheezy dhikata hai.

advantage of notebook on vscode

```
- a = 2+ 3

  a
```

yha pa humy print ka function nai lgaana prta simply wesy hi cell run ho jata hai

ap jitny bar b kernel ko restart krein ga utny bar hi code dubara execute ho jaye ga.

2 bar dd dabany sa cell remove ho jaata hai

cell agar blue ho tou cell ko abhy hum na edit nai kia or jab green ho tou wo running stage ma hota hau esc sa cell blue ho jaye ga cell pa click kry ga tou editing stage ma ajye ga

b dabe ka cell new ban jata hai

yha pa hum cell by cell kam krty hein

cell upar nechy b kar skty hein

```
>print("hello, world")
>
>print(2+3)
>
>print("that was my first program")
```

ctrl enter kia tou sbhy print ho gai

comment wala scene wesy hi hai

restart krny sa cell ki sequence dubara reset ho jaty hau

heading markdown ki trha hoty hai # sa


```
>print("hello, world")
>
>print(2+3)
>
>print("that was my first program")
```


LECTURE 07

indexing in python;

```
a="somosa pakora"
```

```
a

'somosa pakora'
```

is ma humy print ka function use nai krna prta

```
a[5]
```

PRINT :

```
'a'
```

THEN :

```
a[1]
```

PRINT :

```
'o'
```

###another string

```
name="my name is uzair"

name
```

'my name is uzair'

```
name[0:10]
```

'my name is'

```
len(name)
```

16

```
name[-5:12]
```

'u'

```
name.capitalize()
```

'My name is uzair'

```
name.lower()
```

'my name is uzair'

```
name.replace("my", "mein")
```

'mein name is uzair'

##how to split a string

```
fruit="mangoes,apple,grapes,promagrates"

fruit
```

'mangoes,apple,grapes,promagrates'

```
fruit.split(",")
```

['mangoes', 'apple', 'grapes', 'promagrates']

LECTURE 08

data structure in python;

---

tuple

list

dictionaries

set

1-tuple

- ordered collection of elements
- enclosed in ( ) round braces/parenthensis
- different kind of element can be stored
- once elements can stored you can not change them(unmutateable)

```
tup1=(1, "python", True,2.5)

tup1
```

(1, 'python', True, 2.5)

```
#type of tuple

type(tup1)
```

tuple

_ indexing in tuple

```
tup1[1]
```

'python'

```
tup1[0]
```

1

```
#last element is exclusive

tup1[0:3]
```

(1, 'python', True)

counts of elements in tuple

```
len(tup1)
```

4

```
tup2=(2, "papa", 3.5, "false")

tup2
```

(2, 'papa', 3.5, 'false')

concatinate when we add tuple

```
tup1 + tup2
```

(1, 'python', True, 2.5, 2, 'papa', 3.5, 'false')

```
#concatinate + repeat tup1*2 + tup2
```

(1, 'python', True, 2.5, 1, 'python', True, 2.5, 2, 'papa', 3.5, 'false')

```
tup3=(20, 30,40,50)

tup3
```

(20, 30, 40, 50)

```
min(tup3)
```

20

```
max(tup3)
```

50

```
tup3*2
```

(20, 30, 40, 50, 20, 30, 40, 50)

2-list

- ordered collection of elements

- enclosed in square braces[]
- mutateable

```
list1=[2, "papa", 3.5]

list1
```

[2, 'papa', 3.5]

```
type(list1)

list
```

```
len(list1)
```

3

```
list1[2]
```

3.5

```
list2=[3,5,"uzair","what",555,8.98,"true"]

list2
```

[3, 5, 'uzair', 'what', 555, 8.98, 'true']

```
list1 + list2
```

[2, 'papa', 3.5, 3, 5, 'uzair', 'what', 555, 8.98, 'true']

```
list1*2 + list2
```

[2, 'papa', 3.5, 2, 'papa', 3.5, 3, 5, 'uzair', 'what', 555, 8.98, 'true']

```
list1.reverse( )

list1
```

[3.5, 'papa', 2]

```
list2.append("what the hell it will take to much time")

list2
```

[3, 5, 'uzair', 'what', 555, 8.98, 'true', 'what the hell it will take to much time']

```
list3=[20,45,67,98,90,99]

list3
```

[20, 45, 67, 98, 90, 99]

```
len(list3)
```

6

```
list3.sort()

list3
```

[20, 45, 67, 90, 98, 99]

3-dictionaries

- an unordered collectiin of elements
- key and value
- curly braces/ or brackets
- mutateable

#food and thier prices

```
food1 = {"samosa":30, "pakora":100,"raita":10, "salad":40}

food1
```

{'samosa': 30, 'pakora': 100, 'raita': 10, 'salad': 40}

#extract food

```
keys1 = food1.keys()

keys1
```

dict_keys(['samosa', 'pakora', 'raita', 'salad'])

```
value1 = food1.values()

value1
```

dict_values([30, 100, 10, 40])

```
food1["Tikki"]=10

food1
```

{'samosa': 30, 'pakora': 100, 'raita': 10, 'salad': 40, 'Tikki': 10}

#update the value

```
food1["Tikki"]=15

food1
```

{'samosa': 30, 'pakora': 100, 'raita': 10, 'salad': 40, 'Tikki': 15}

```
food2 = {"dates": 50, "chocolates": 200, "sawwaya": 1000}

food2
```

{'dates': 50, 'chocolates': 200, 'sawwaya': 1000}

#concatinate

```
food1.update(food2)

food1
```

{'samosa': 30, 'pakora': 100, 'raita': 10, 'salad': 40, 'Tikki': 15, 'dates': 50, 'chocolates': 200, 'sawwaya': 1000}

4-sets

- unordered and unindexed
- curly braces[]
- no duplicate allowed

```
s1=[1, 2, 3.5, "uzair", "cold joke"]

s1
```

[1, 2, 3.5, 'uzair', 'cold joke']

```
s1.append("uzair2")

s1
```

[1, 2, 3.5, 'uzair', 'cold joke', 'uzair2']

```
s1.remove("uzair2")

s1
```

[1, 2, 3.5, 'uzair', 'cold joke']

LECTURE 09

live session;

- body mass index (BMI)
- input weight
- height
- BMI (ask yourself your height)
- weight
- calculate BMI
- print (my name is my height is ....)

- your weight in kilogram / your height in meter and than its square

```
height = input("what is your height? ")

height
```

what is your height? 1.74

'1.74'

```
height = float(height)
```

yha pa type casting ki hai

```
weight = input("what is your weight? ")

weight
```

what is your weight? 68

'68'

```
weight = float(weight)
```

```
name = input("what is your name? ")

name
```

what is your name? uzair

'uzair'

```
BMI = (weight/height**2)

BMI
```

22.460034350640772

```
print("my name is", name, "AND my BMI is", BMI)
```

my name is uzair AND my BMI is 22.460034350640772

---

fundamentals of visualization

graphic system for language r

codes to graphes

code ko graph ma convert krna hai ap na batana ya hmara data hai or ya hmra code hai ya graph hai ap ap na aik triangle banany hai ap wo kis tariqy sa bnaty hein data code graph ya 3 cheezy is o kis tariqy sa bnaty hein isy data visualization khty hein.

minds to page and plots

page and plot ma mind banana hai

what is grammer of graphic?

1. mapping(data) like jesy ballon board ko gun sa ballon sa kuch bna dein
2. aesthetic(color,shape,size)
3. geometric(objects: line ,bar,points,box,map) different types ka plot hoty hein.

may also contain minor parts-data on specific cordinates

baba ammar theory

work like an artist or be a copy cat

libraries in python for data visualization

- matplotlib(plot graphs and figure)
- seaborn(heat maps,time series and other plots)

Machine learning algorithmic development libraries

- scikit-learn
- statsmodels

variables type matter

type of visualization depend on the variable type

- categorical variable

1. counts plot type
2. male vs female
3. true vs false
4. 0 vs 1
5. yes or no jesy pole bnaty heun

- continuous variable

1. scatter plot
2. statistical proportions

LECTURE 10

live session;

---

cmd ma absolute directory dein cd.. sa aik directory pechy chly jaty hein cd. ma usy directory ma rhty hein new directory bnany ka liye mkdir likh KAR AGY US KA NAME da dein diresctory check krny ka liye dir likhy tap ka mtlb programming ma auto complete hota hai agy libraries install karwaty hein

_steps invovled in data visualization _

step 1 import librarie like

import seaborn as sns

step2 set a theme

sns.set_theme(style="ticks", color_codes=true)

step3 import data set you can also import your own data set

kashti=sns.load_dataset("titanic")

print(kashti)

step4 plot basic graph with 1 variable (count)

p = sns.countplot(x="sex ",data=kashti) plt.show()

step4 plot basic graph with 2 variable (count plot) with title

p = sns.countplot(x="sex ", ,data=kashti, hue="class")

p.set_title("title name")

plt.show()

LECTURE 11

line and bar plot;

---

import libraries

seaborn automatically install these libraries

- numpy
- spicy
- pandas
- matplotlib

#import libraries

```
import seaborn as sns

import matplotlib.pyplot as plt
```

in code mein hum na libraries import ki hein

#import dataset

```
phool=sns.load_dataset("iris")

phool
```

Index	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa

Index	sepal_length	sepal_width	petal_length	petal_width	species
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

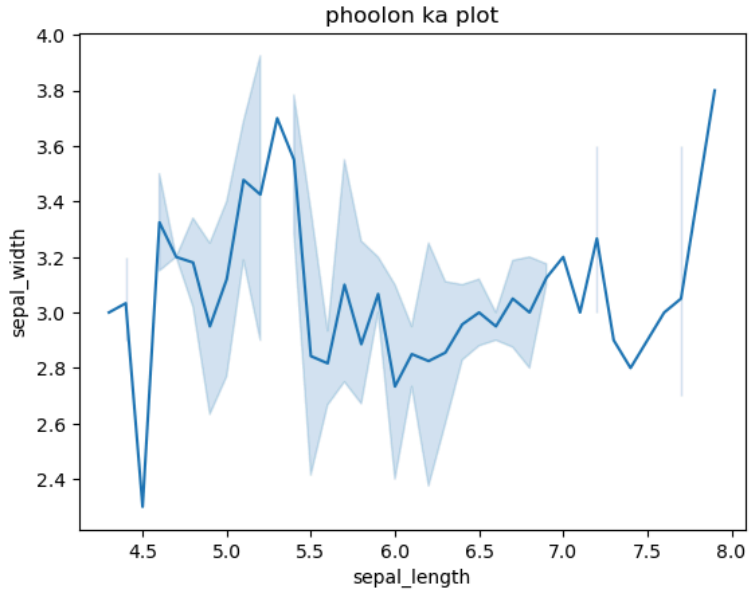
in lines mein hum na phool variavle rkha agy library sa data import kia hai

#draw a line plot

```
sns.lineplot(x="sepal_length", y="sepal_width", data=phool)

plt.title("phoolon ka plot")

plt.show
```



in lines mein humna plot bnany ka liye data diya phr title dia phir output ka la function dia

##adding limits

```
import seaborn as sns

import matplotlib.pyplot as plt phool=sns.load_dataset("iris")

phool

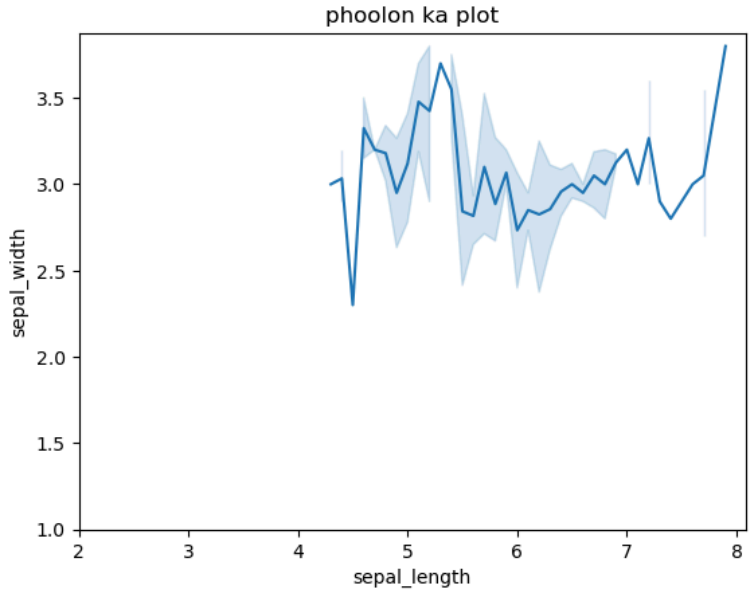
sns.lineplot(x="sepal_length", y="sepal_width", data=phool)

plt.title("phoolon ka plot")

plt.xlim(2)

plt.ylim(1)

plt.show
```



##set styles

- darkgrid
- whitegrid
- dark
- white
- ticks

```
import seaborn as sns

import matplotlib.pyplot as plt

sns.set_style(style=None, rc=None)

phool=sns.load_dataset("iris")

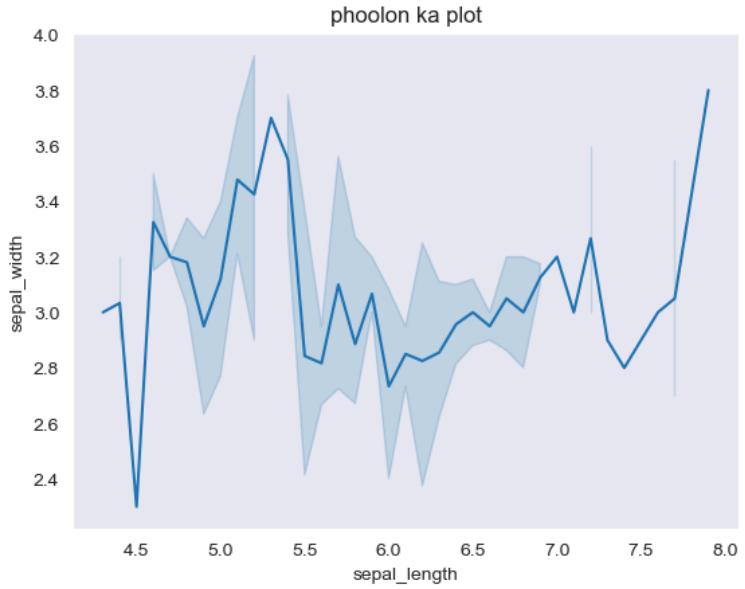
phool

sns.lineplot(x="sepal_length", y="sepal_width", data=phool)

plt.title("phoolon ka plot")

sns.set_style("dark")

plt.show
```



##size of figure

```
import seaborn as sns

import matplotlib.pyplot as plt
```

```
phool=sns.load_dataset("iris")

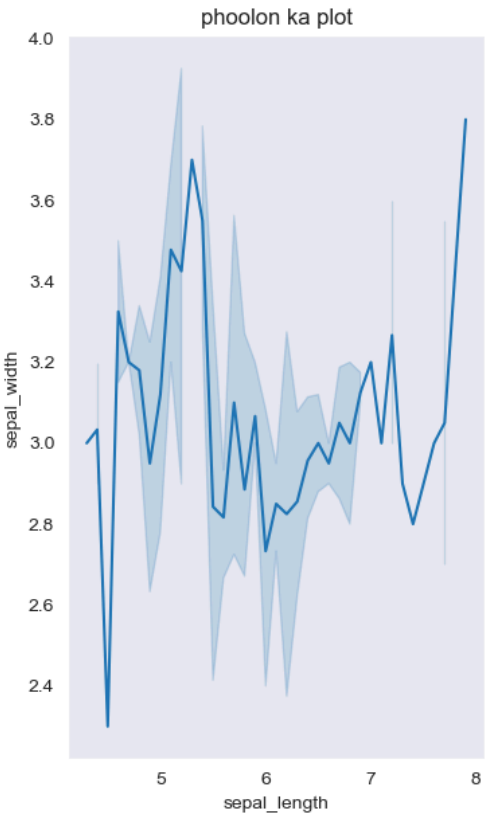
#change figure

plt.figure(figsize=(4,7))

sns.lineplot(x="sepal_length", y="sepal_width", data=phool)

plt.title("phoolon ka plot")

plt.show
```



LEXTURE 13

boxplot in python

```
import seaborn as sns

import matplotlib.pyplot as plt

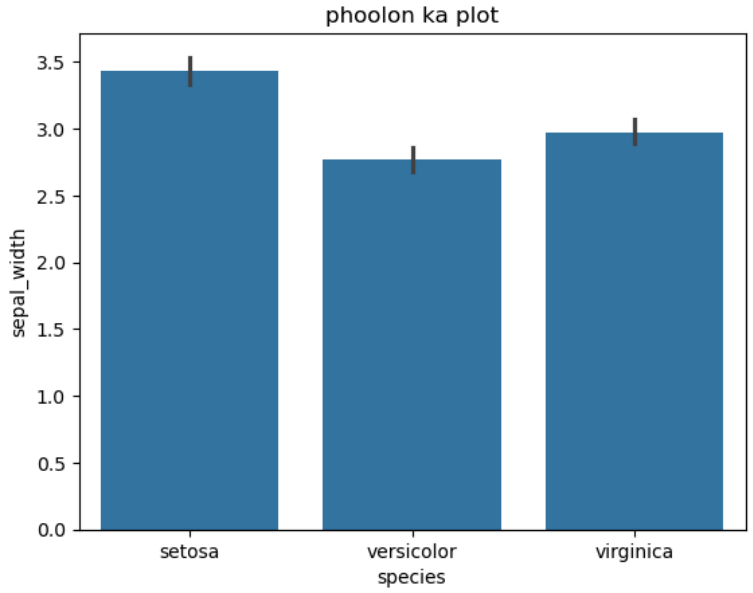
phool=sns.load_dataset("iris")

phool

sns.barplot(x="species", y="sepal_width", data=phool)

plt.title("phoolon ka plot")

plt.show
```



phool

Index	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

```
import seaborn as sns

import matplotlib.pyplot as plt

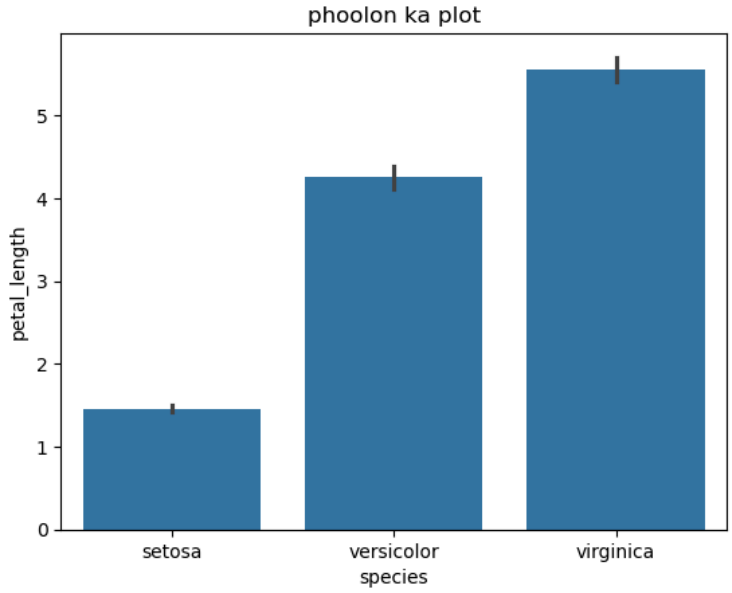
phool=sns.load_dataset("iris")

phool

sns.barplot(x="species", y="petal_length", data=phool)

plt.title("phoolon ka plot")

plt.show
```



#grouping

```
import seaborn as sns

import matplotlib.pyplot as plt

kashti=sns.load_dataset("Titanic")

kashti
```

Index	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
886	0	2	male	27.0	0	0	13.0000	S	Second	man	True	NaN	Southampton	no	True
887	1	1	female	19.0	0	0	30.0000	S	First	woman	False	B	Southampton	yes	True
888	0	3	female	NaN	1	2	23.4500	S	Third	woman	False	NaN	Southampton	no	False
889	1	1	male	26.0	0	0	30.0000	C	First	man	True	C	Cherbourg	yes	True
890	0	3	male	32.0	0	0	7.7500	Q	Third	man	True	NaN	Queenstown	no	True

```
import seaborn as sns

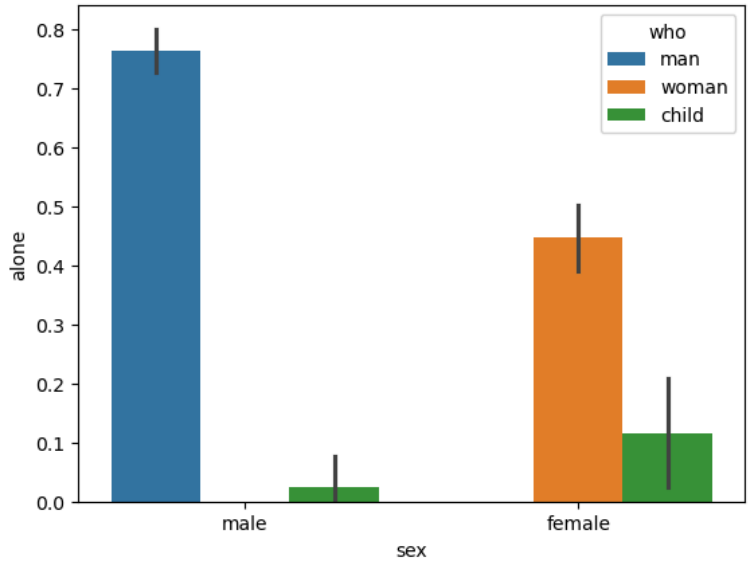
import matplotlib.pyplot as plt

kashti=sns.load_dataset("titanic")

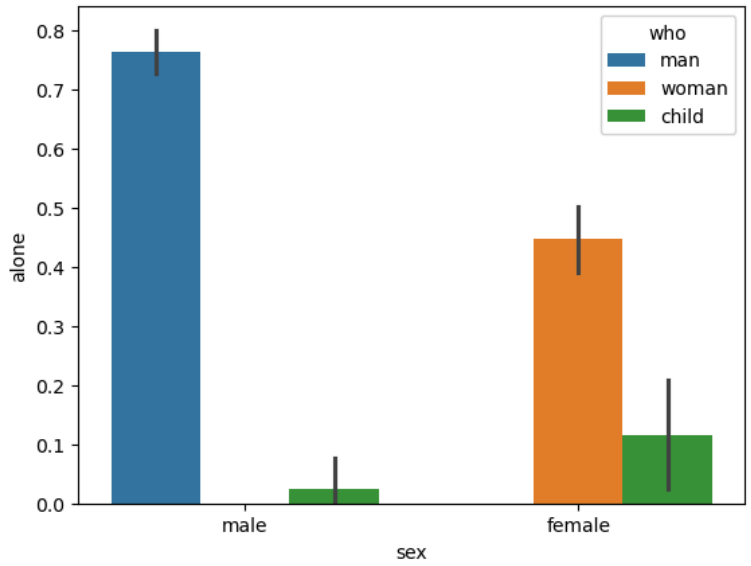
kashti

sns.barplot(x="sex", y="alone", hue="who", data=kashti)

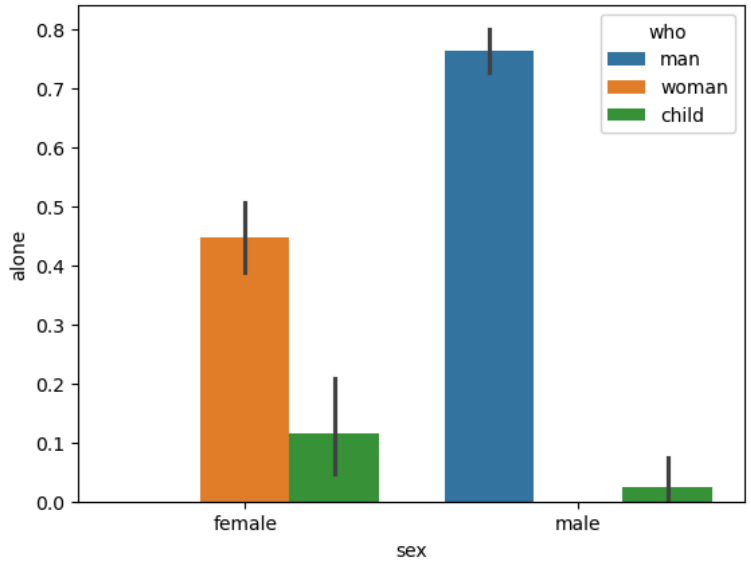
plt.show
```




```
import seaborn as sns
import matplotlib.pyplot as plt
kashti=sns.load_dataset("titanic")
kashti
sns.barplot(x="sex", y="alone", hue="who", data=kashti)
plt.show
```

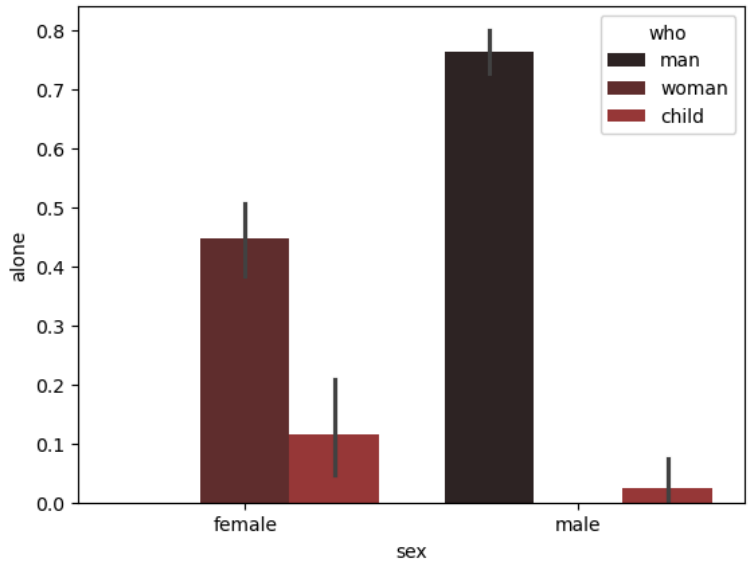



```
import seaborn as sns
import matplotlib.pyplot as plt
kashti=sns.load_dataset("titanic")
kashti
sns.barplot(x="sex", y="alone", hue="who", data=kashti, order=["female", "male"])
plt.show
```

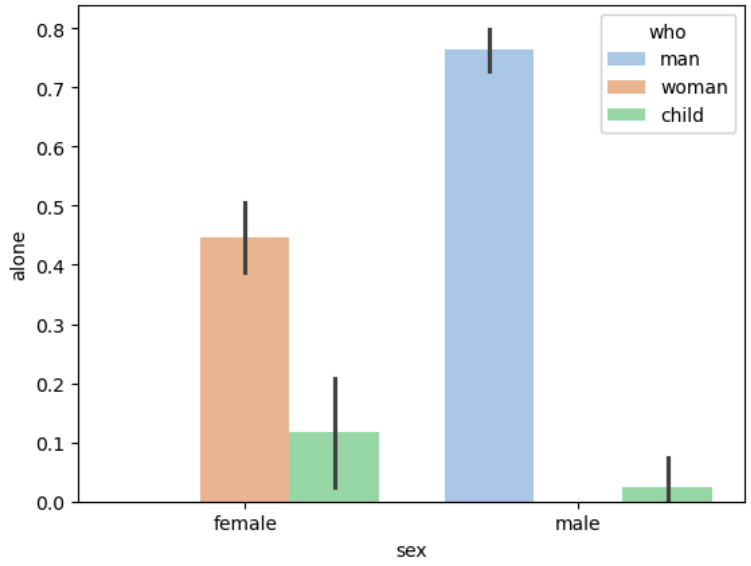


##change color

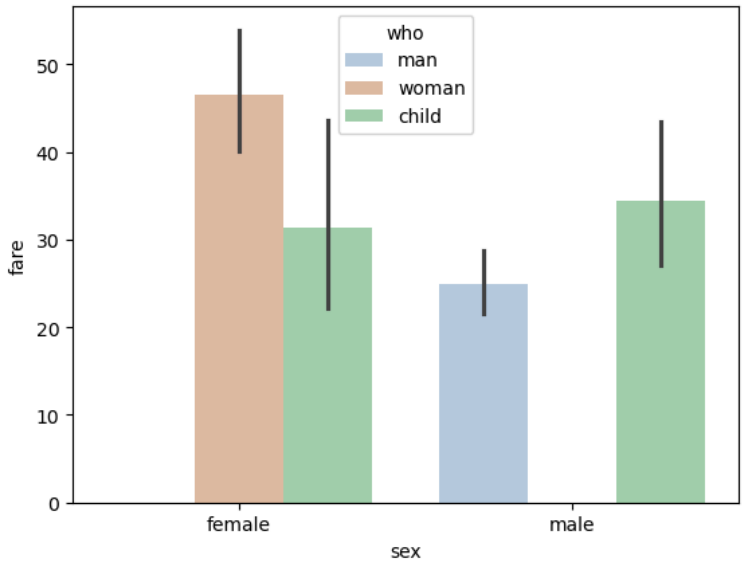
```
import seaborn as sns
import matplotlib.pyplot as plt
kashti=sns.load_dataset("titanic")
kashti
sns.barplot(x="sex", y="alone", hue="who", data=kashti, order=["female", "male"], color="brown")
plt.show
```




```
import seaborn as sns
import matplotlib.pyplot as plt
kashti=sns.load_dataset("titanic")
kashti
sns.barplot(x="sex", y="alone", hue="who", data=kashti, order=["female", "male"], color="brown", palette='pastel')
plt.show
```

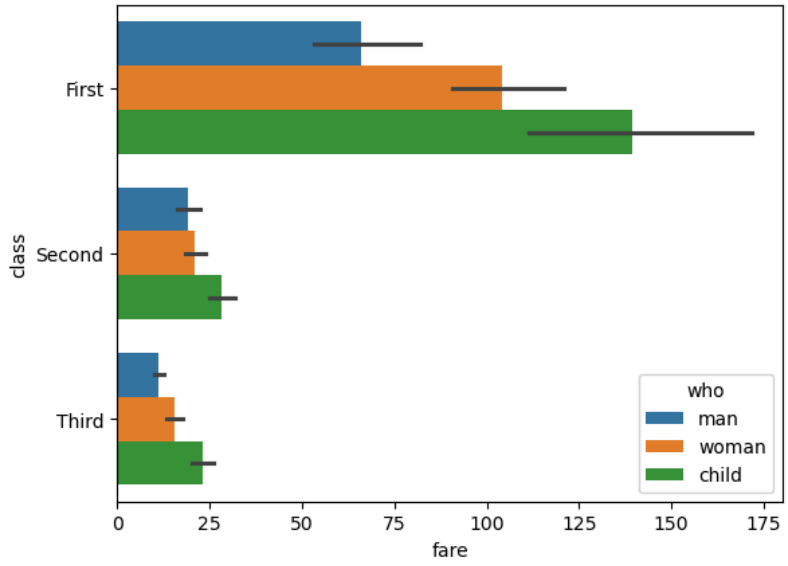



```
import seaborn as sns
from numpy import mean
import matplotlib.pyplot as plt
kashti=sns.load_dataset("titanic")
kashti
sns.barplot(x="sex", y="fare", hue="who", data=kashti, order=["female", "male"], color="brown", palette='pastel',estimator=mean,
saturation=0.5)
plt.show
```



#horizontal plot

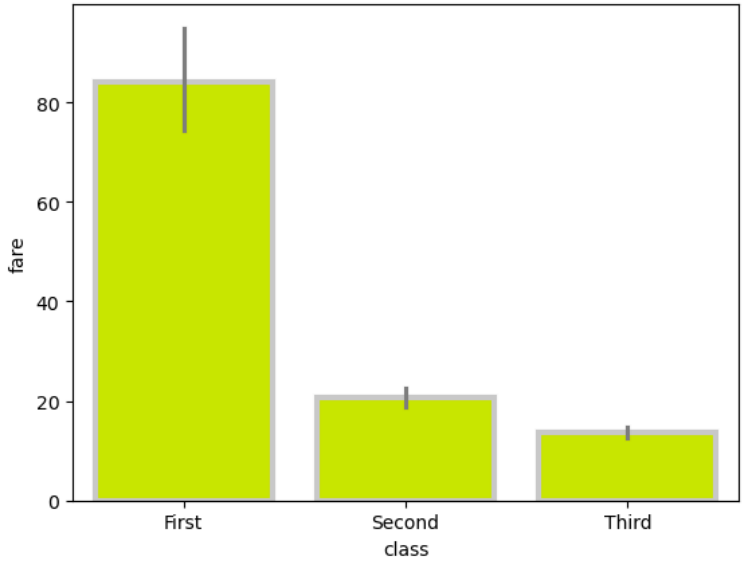
```
import seaborn as sns
from numpy import mean
import matplotlib.pyplot as plt
kashti=sns.load_dataset("titanic")
kashti
sns.barplot(x="fare", y="class", hue="who", data=kashti)
plt.show
```




```
#importing library
import seaborn as sns
import matplotlib.pyplot as plt

#read a titanic.csv file
#from seaborn library
kashti=sns.load_dataset("titanic")

sns.barplot(x="class", y="fare", data=kashti,
            linewidth=3, facecolor=(.8, .9, 0),
            errcolor=".5", edgecolor=".8")
```



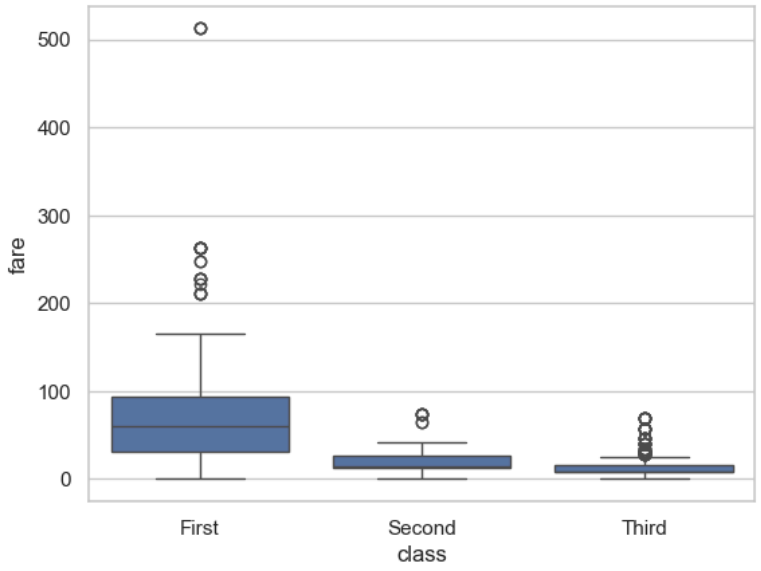
LECTURE 14

customizing plot in python;

```
#import seaborn libraray
import seaborn as sns
#canvass (ballon board)
sns.set(style="whitegrid")

kashti=seaborn.load_dataset("titanic")

sns.boxplot(x="class", y="fare", data=kashti)
```




```
import seaborn as sns
sns.set(style="whitegrid")
tip=sns.load_dataset("tips")
tip
```

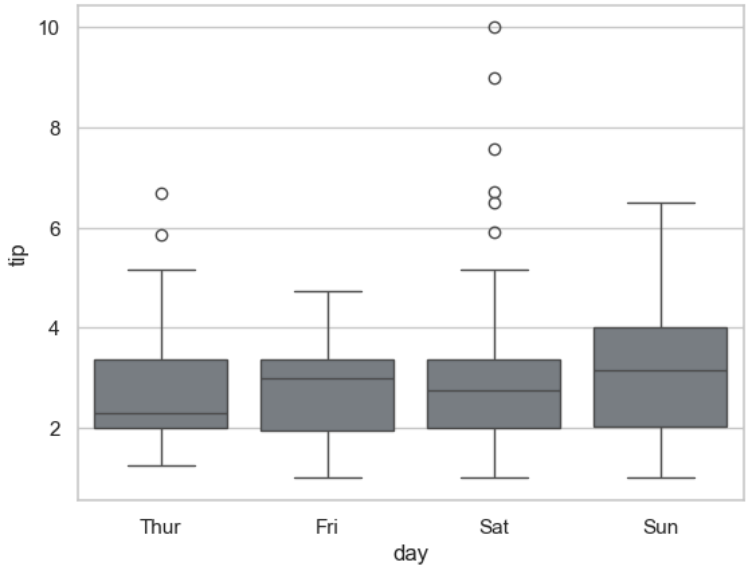
Index	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

```
import seaborn as sns
import pandas as pf
import numpy as np

tip=sns.load_dataset("tips")
tip.describe
```

Statistic	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

```
import seaborn as sns
sns.set(style="whitegrid")
tip=sns.load_dataset("tips")
sns.boxplot(x="day", y='tip', data=tip, saturation=0.1)
```




```
import seaborn as sns
import pandas as pf
import numpy as np
tip=sns.load_dataset("tips")
tip
```

Index	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

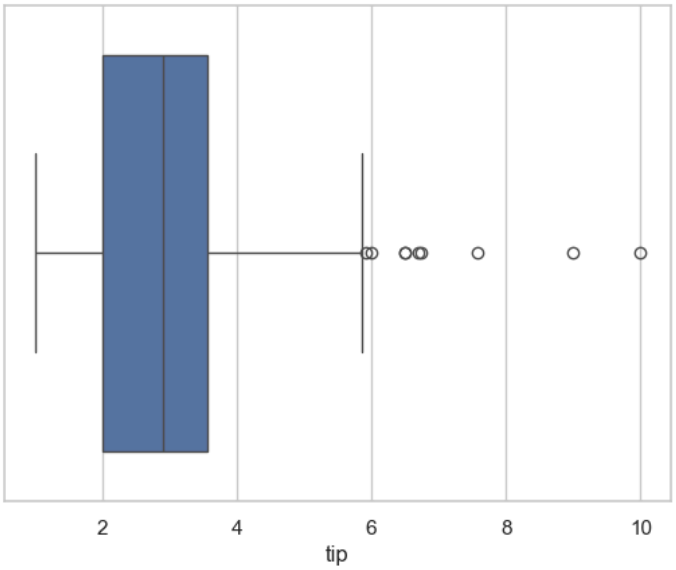
tip.describe()

Statistic	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

```
import seaborn as sns
# use to set style of background of plot
sns.set(style="whitegrid")

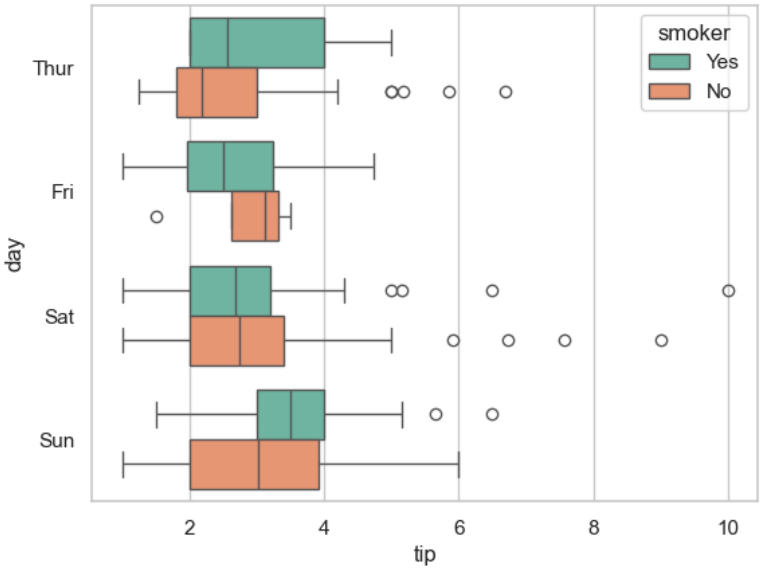
tip=sns.load_dataset("tips")

sns.boxplot(x=tip['tip'], )
```




```
import seaborn as sns
# use to set style of background of plot
sns.set(style="whitegrid")

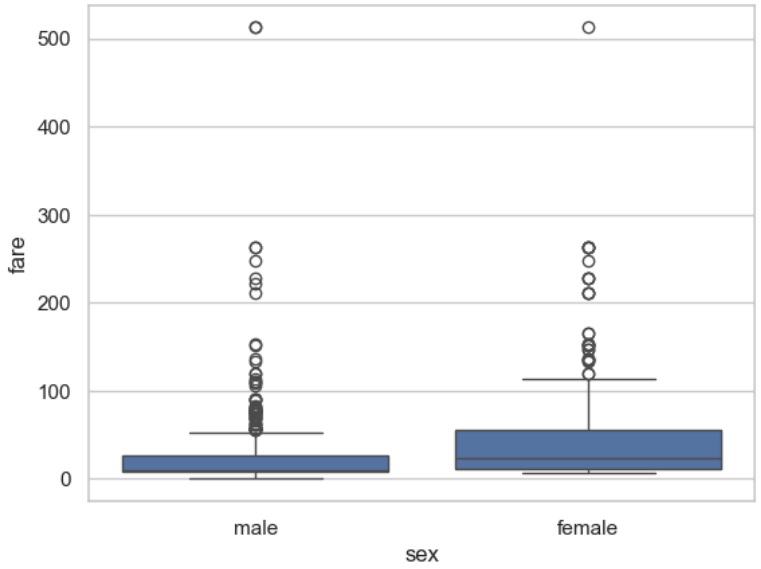
tip=sns.load_dataset("tips")
sns.boxplot(x='tip', y='day',hue='smoker', data=tip,
            palette="Set2", dodge=True)
```




```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pf
import numpy as np
kashti=sns.load_dataset("Titanic")
kashti.head()
```

Index	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

```
sns.boxplot(x="sex",
            y="fare", data=kashti)
```




```
sns.boxplot(x="survived",
            y="age", showmeans=True,
            meanprops={"marker":"*",
                       "markersize":"12",
                       "markeredgecolor":"red"},
            data=kashti)
# show labels
plt.xlabel("how many survived", size = 12, weight='bold')
plt.ylabel("age (years)", size= 14, weight='bold')
plt.title("box plot of kitny doobay or kitnay bach gye",size = 16, weight='bold')

plt.show()
```

